Cloud Computing

Introduction

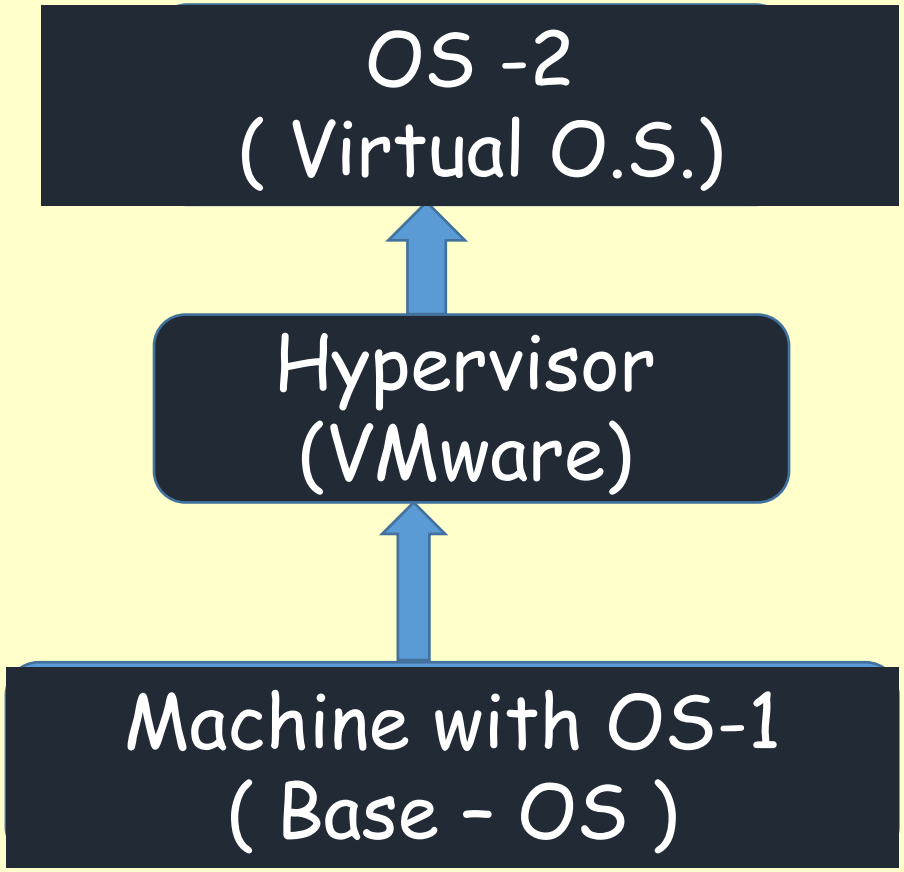
- **Internet** and **Intranet**
- **Cloud** – higher level internet. Cloud is internet or intranet.
- Background to **Cloud Computing** –
  - ✓ The **transfer speeds** of the internet and connectivity got better and it gave way to new type of companies called **Application Service Providers (ASPs)**.
  - ✓ ASPs took the existing business applications and ran them for the business using their own machines.
  - ✓ The customers would pay a monthly fee to run their business over the internet from ASP's systems.

Why Cloud Computing

- Advantage over on–premise net setup.
- Pay as per use per second/minute/hour basis
- No capital expenditure
- Getting new server is as easy as clicking of mouse buttons
- Provides very high available solutions without any effort
- Security
- Flexibility and Error shield

Virtualization

- **Virtualization** is creation of virtual version of something like **OS**, **server, storage device** or **network resources**.



Hypervisor

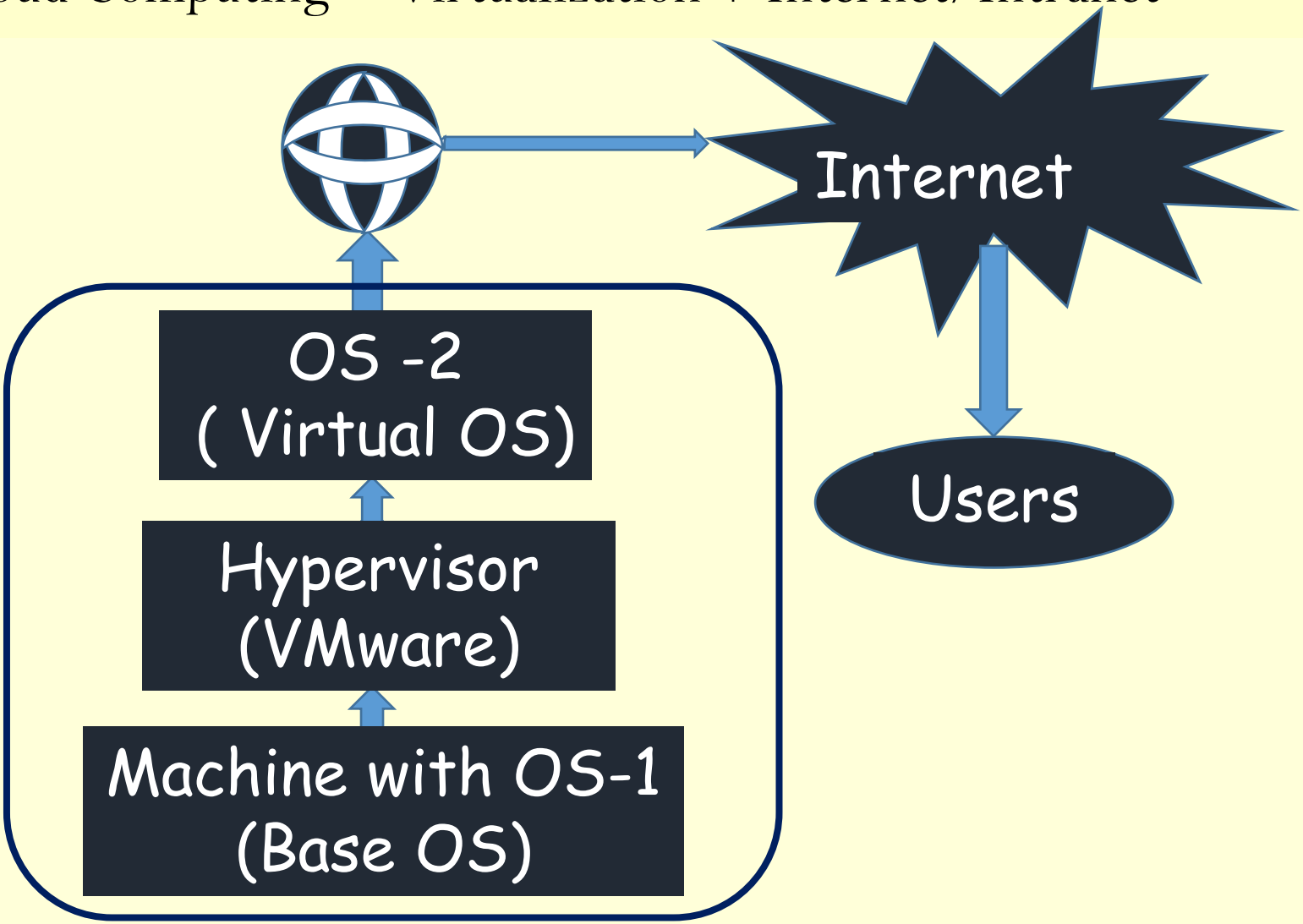
- **Hypervisor** makes it easy to manage **multiple** operating systems in a **single host system**.
- The **independence** and **co-existence** of different operating systems is made possible by hypervisor.
- Security offered to the systems by hypervisor should be appreciated against both external and internal virus attacks.
- Free hypervisors are available in the market.
- Two types

  Type 1: Native hypervisor. Hosted on top of the hardware.

  Type 2: Hosted hypervisor. Its hosted on top of the OS.
- Hypervisor used by AWS is XEN Hypervisor. KVM is also used as hypervisor. (Kernel Virtual Machine).

Cloud Computing

- Virtualized computing environment on internet/intranet
- Cloud Computing = Virtualization + Internet/Intranet



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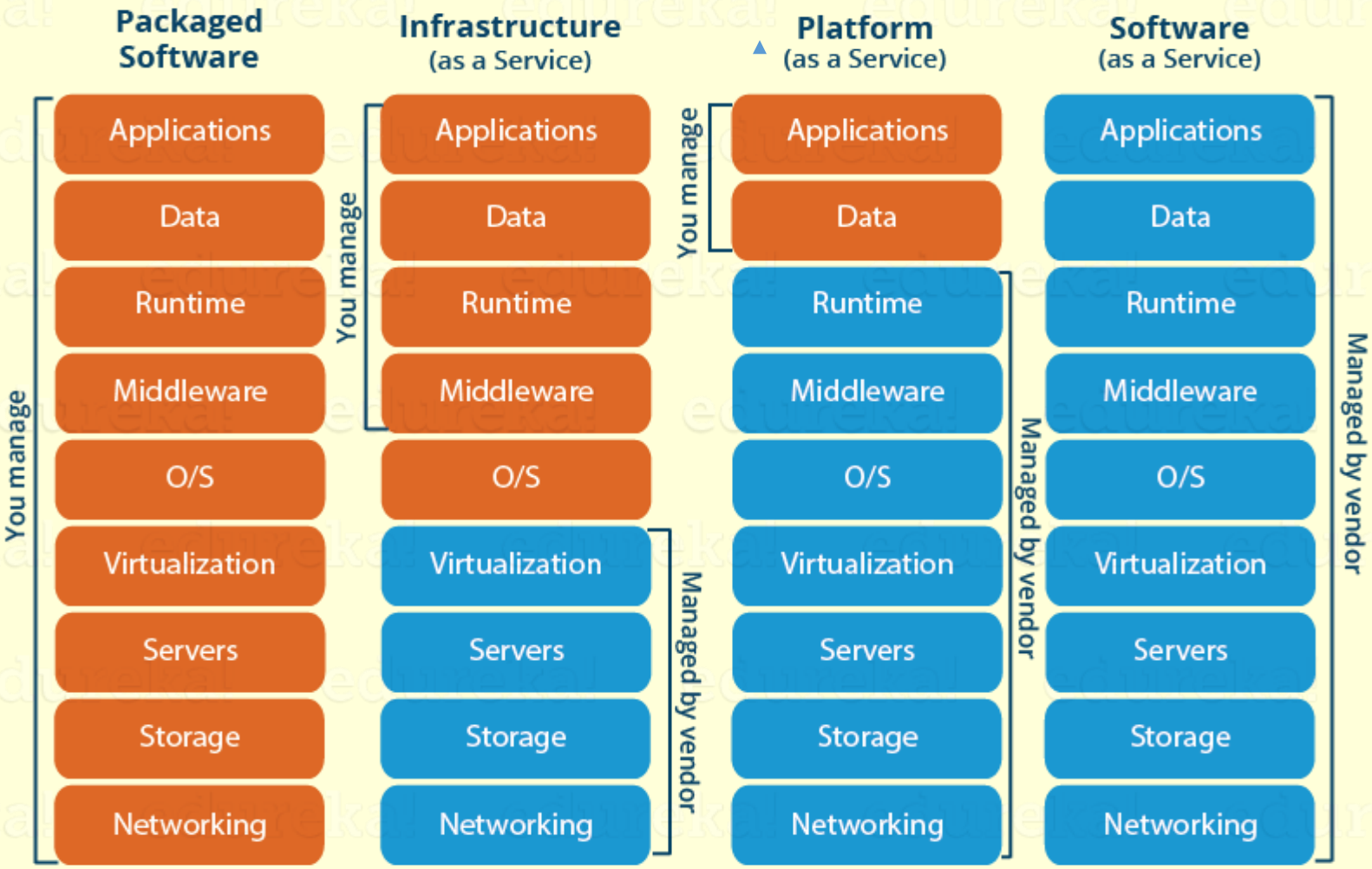
- **Cloud Computing** refers to **computing over internet**, a means to store/access **user data** and **applications** over internet.
- A **model** delivering **on-demand, self-sufficient–services** like viz.. Virtualization, Storage, Network, Operating System, Middleware, Databases, Security, Applications.
- Users utilize these services with little or no interaction with service providers.
- The major companies delivering the Cloud Computing Services –

1. **Amazon**: **Amazon Web Service(AWS)** is one of the best Cloud Computing Service provider which offers a wide set of infrastructure services like database storage, computing power, networking etc.
2. **Google**: **Google Cloud Platform** allows clients to build, test, and deploy applications on Google's highly-scalable and reliable infrastructure.
3. **Microsoft**: **Microsoft Azure** is used for deploying, designing and managing the applications through a worldwide network.

Service Models

- Iaas – AWS, Rackspace, MS Azure
- Paas – AWS Elastic, Beanstalk, Heroku, etc
- Saas – Google Drive, Google docs, MS Office 365 etc.



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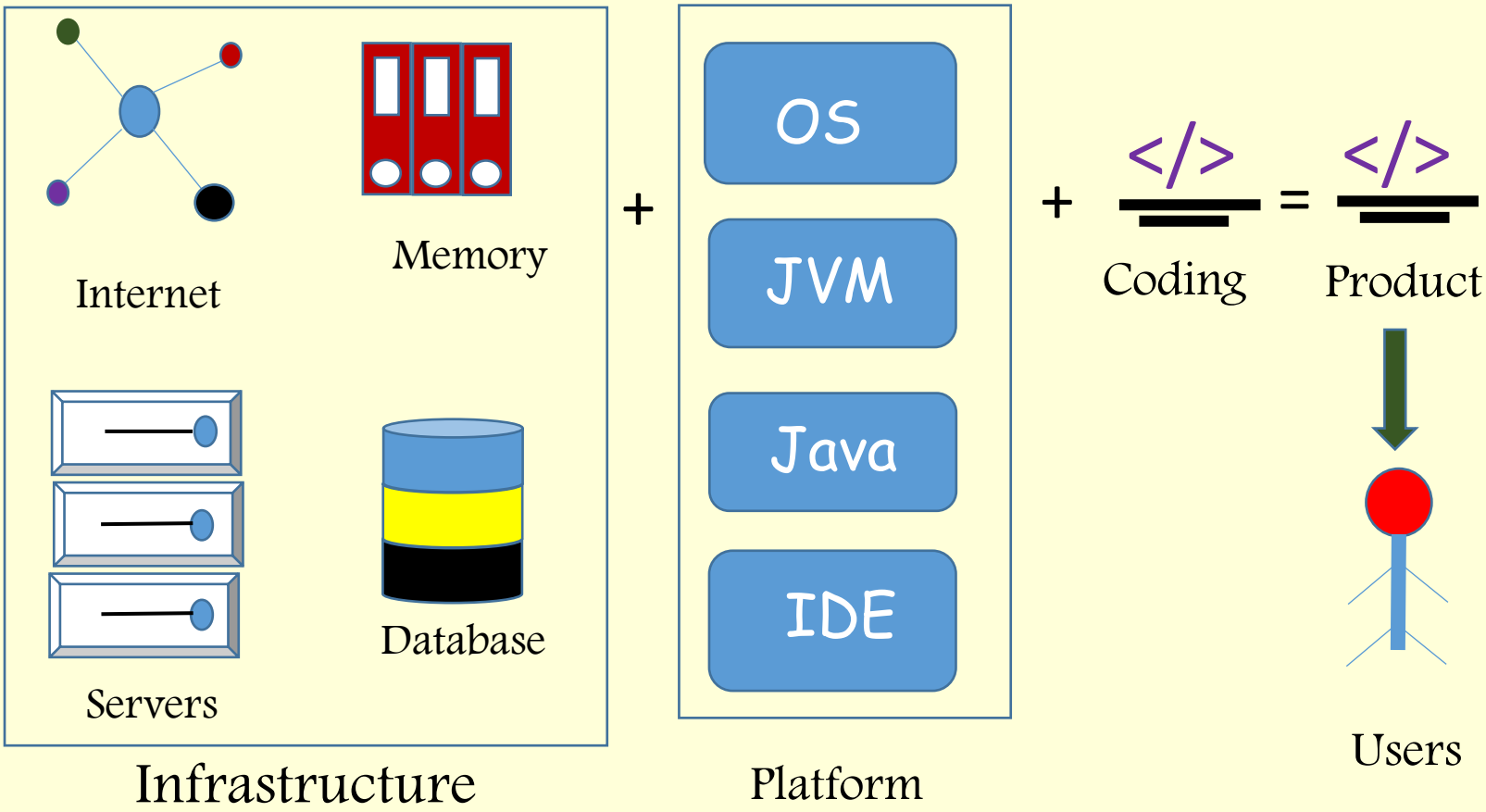

Cloud deployment models

- **Public Cloud** – In **public cloud** deployment mode, the services which are deployed are open for public use and generally public cloud services are free. The difference between a public cloud and a private cloud are the security parameters are very different.

  AWS, MS Azure, Google Platform, IBM Bluemix

- **Private Cloud** – A **private cloud** is operated solely for a single organization, can be done by self or a third-party organization. The costs are high since the hardware should be updated periodically, security also has to be kept in check since new threats come up daily.

  HPE, VMWare, RedHat, OpenStack, Dell, EMC etc

- **Hybrid** – A hybrid cloud consists the functionalities of both private and public cloud.

Life cycle of Cloud computing solution

- To create such a cloud platform takes a long number of steps and dedicated time.

Step 1: Define the Purpose - The first and foremost step is to **define the purpose** for which to create a cloud. Involves understanding business requirement, type of application to run on the cloud, decide on type – public, private, or hybrid.

Step 2: Define the Hardware - Decide precisely what **hardware is needed** in the next setup. Need to choose the **compute service** that provides the right support when resize compute capacity to maintain the application running.

Step 3: Define the Storage – Any **application storage type** that should be chosen carefully for this, one should choose the storage service where back up and archive their data over the internet is safe.

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Step 4: Define the Network – The network configured should be flawless so that intruders can not break into the network. One should define the network that securely delivers data, videos, and applications with low latency and high transfer speed.

Step 5: Define the Security – Security is a key aspect of any application. Set up of security service involves enabling services for user authentication or limiting access to a certain set of users on the resources.

Step 6: Define the Management process and tools – The developer must have complete control over the resource and to configure these define some management tools which monitor the cloud environment, resources used, and the customer application running on it.

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Step 7: Testing the Process – Testing is yet another important thing in the life cycle of deploying any application. All the faults can be figured out only through the testing process involved in it.

Step 8: Analytics – Finally, analyze and visualize data using analytics service where data can be queried instantly and get results then and there only. Once analyzing is done complete, application becomes ready to deploying.

Service level agreement (SLA)

- Contract between Service Provider and Customer that defines the level of service expected from the service provider.
- If SLA is missed, penalty can be levied on that Service Provider.
- Every service of AWS must have an SLA.
- E.g. SLA of EC2 : UPTIME SLA for a server in Corporate DC

SLA	**Total(mins)**	**Doowntime(mins)**
95%	43200	2160
99%		432
99.9%		43.2
99.99 %		4.32

Amazon Web Services

- **Amazon Web Services(AWS)** is a **Amazon.com** subsidiary which offers **cloud – computing services** at affordable rates and making its customer base strong from small scale companies to big enterprises.

- The world's most comprehensive and widely used cloud platform, delivers around 240 fully-featured services from data centers all over the world (number as of 2026).
- Amazon Web Services' talents and knowledge are in great demand, and may continue to be so in the future.
- AWS certification increases the pay range or quality needed. It boosts credibility, which will help to advance in job.

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- Some of the different domains in which AWS offer services –

- **Compute** used to process data on the cloud by making use of powerful processors which serve multiple instances at a time.
- **Storage and Content Delivery** used to store data in the cloud; can be stored anywhere but content delivery is used to cache data nearer to the user so as to provide low latency.
- **Database** used to provide reliable relational and non relational database instances managed by AWS.
- **Networking** includes services which provide a variety of networking features such as security, faster access etc.
- **Management Tools** includes services which can be used to manage and monitor your AWS instances.
- **Security and Identity** includes services for user authentication or limiting access to a certain set of audience on AWS resources.
- **App Services** includes services e.g notifications, email, queuing.

Categorization of Services

- **Compute**

AWS EC2 is a web service which provides re-sizable compute capacity in the cloud. It is designed to make the web scale computing easier for developers.

- **AWS Elastic Beanstalk**

Elastic Beanstalk provides to quickly deploy and manage applications in AWS without worrying about the underlying infrastructure.

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- **AWS Lambda**

  AWS Lambda is used to execute backend code without worrying about the underlying architecture, you just upload the code and it runs, it's that simple.

- **AWS Elastic Load Balancing**

  ELB automatically manages the workload on your instances and distributes them to other instances in case of an instance failure.

- **AWS Autoscaling**

  The Autoscaling feature is used to scale up and down automatically as and when required.

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- **Storage and Content Delivery**
- ✓ **S3 AWS**

S3 stands for simple storage service, it is used for storing data in the form of objects in the AWS Cloud.

- **Amazon CloudFront**

- ✓ CloudFront is a content delivery network which is used to cache data to an edge location which reduces latency.
- **Amazon EBS**

- ✓ Amazon Elastic Block Storage is a storage service wherein each block of storage acts like a separate hard drive.

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- **Amazon Glacier**
  Glacier is an archiving service offered by Amazon, which offers low cost data archiving.

- **AWS Import/Export Snowball**
  It offers physical transfer of data between user's location and AWS data centers, the device which is used to transfer the data is called Snowball.

- **AWS Storage Gateway**
  It is used to provide seamless integration with data security features between your on premise software appliance and AWS Cloud.

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- **Database**
- **Amazon Aurora**

It is a relational database engine that combines the speed and reliability of high-end commercial databases and the cost effectiveness and simplicity of open-source databases.

- **Amazon RDS**

Amazon RDS is a managed relational database service which does routine database tasks in familiar databases e.g. MySQL, MariaDB, Oracle, Microsoft SQL Server, and PostgreSQL.

- **Amazon DynamoDB**

It is a fully managed No-SQL database service. It is known for extremely low latencies and scalability.

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- **Amazon ElasticCache**

It is a web service that makes it easy to set up, manage and scale a distributed cache-in environment in the cloud.

- **Amazon Redshift**

Amazon Redshift is a fully managed petabyte-scale data warehouse service in the cloud.

- **Networking**
- **VPC AWS**

Amazon VPC lets you launch AWS resources in a virtual network that you define. It closely resembles a traditional network of data center.

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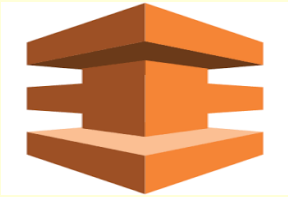
- **AWS Direct Connect**



It helps to establish a private connection between user premises and AWS, therefore giving better network performance and throughput than an Internet based connection.

- **Amazon Route 53**

Route 53 is a highly scalable and highly available Domain Name System by Amazon AWS. The name is in reference to the TCP and UDP's port 53 where DNS requests are addressed.

contd..

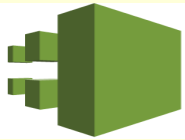
- **Management Tools**
- **Amazon CloudWatch**

It is a monitoring tool by AWS which is used to keep a track on the AWS resources and the applications you run on Amazon AWS.

- **AWS CloudFormation**

It is a service which helps you setup and model your Amazon AWS resources so that you can spend less time managing these resources and more time focusing on the development.

- **AWS CloudTrail**



AWS CloudTrail is a logging service which records the API calls to user's Amazon AWS account.

contd..

- **AWS Command Line Tool**

It is an all-in-one tool to manage all AWS services. By downloading and configuring only one tool all the AWS services can be managed through the command line.

- **AWS OpsWorks**

It is a configuration management tool that helps configure and operate applications of all size and shapes using Chef.

- **Trusted Advisor**

Trusted Advisor is a customized cloud monitoring tool, that analyzes user's AWS environment and gives insights on the expense, performance improvement, security gaps and reliability.

contd..

- **Security and Identity**

- **AWS Identity and Access Management(IAM)**

  It is an AWS service that helps to control access to your AWS resources for your users.

- **AWS Key Management Service**

  It is a managed service that helps to create and control encryption keys which is used to encrypt user data, and uses Hardware Security Modules to protect the security of user keys.

contd..

- **Application Services**

- **Amazon SES**

It is a cost-effective emailing service which is built on the scalable and reliable infrastructure of Amazon.com

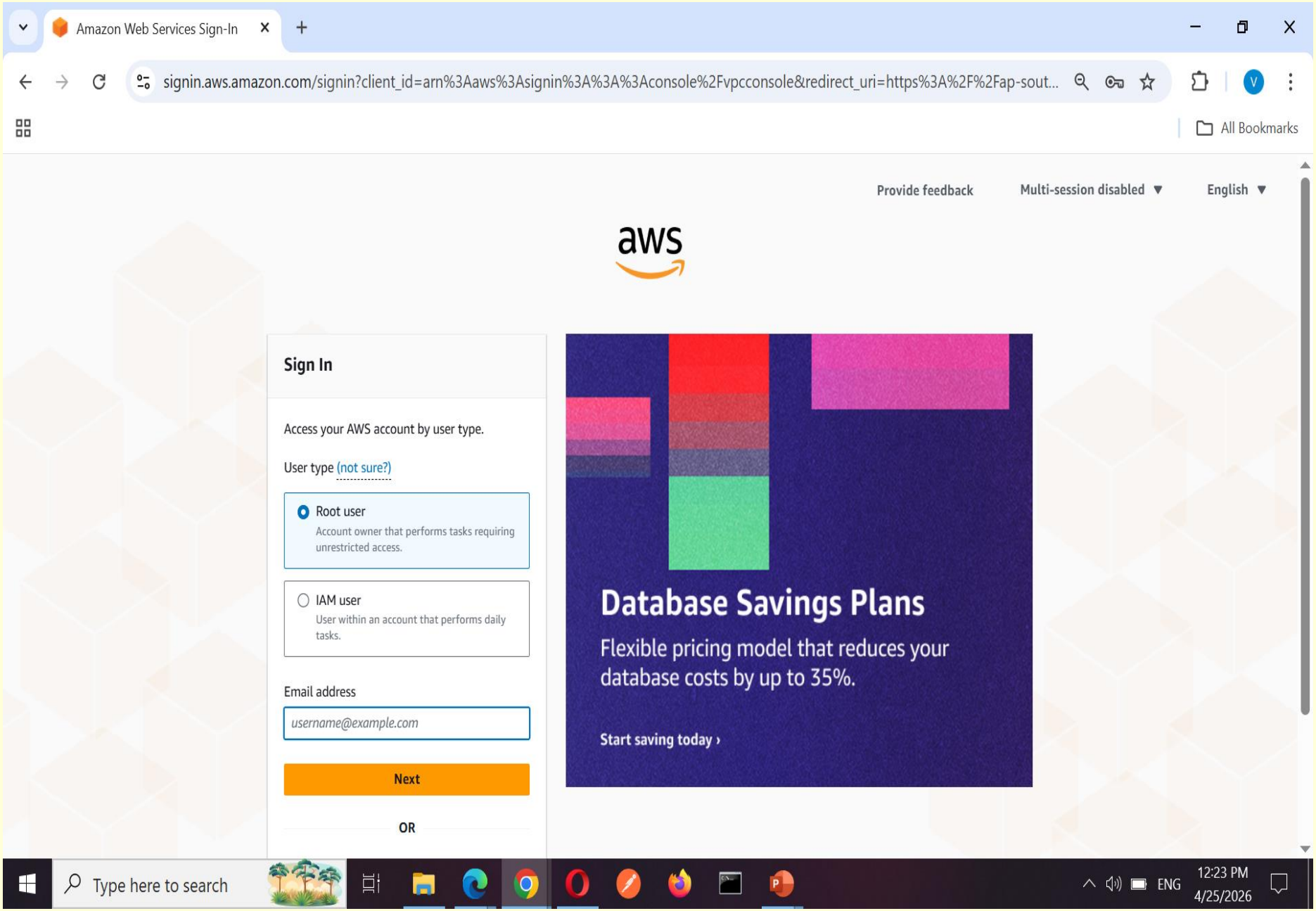
- **Amazon SNS**

It is a web service offered by AWS that manages the delivery of messages to subscribed **endpoints or clients.**

- **Amazon SQS**

It is a fast, reliable and scalable message queuing service, it can be used to transmit any volume of data at any level of throughput, without losing any messages or without the use of any other service.

AWS login


Amazon Web Services Sign-In
signin.aws.amazon.com/signin?client_id=arn%3Aaws%3Asignin%3A%3A%3Aconsole%2Fvpcconsole&redirect_uri=https%3A%2F%2Fap-sout...
All Bookmarks
Provide feedback
Multi-session disabled
English
aws
Sign In
Access your AWS account by user type.
User type (not sure?)
Root user
Account owner that performs tasks requiring unrestricted access.
IAM user
User within an account that performs daily tasks.
Email address
username@example.com
Next
OR
Database Savings Plans
Flexible pricing model that reduces your database costs by up to 35%.
Start saving today ›
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AWS Management Console

All services | Console Home | us

us-east-1.console.aws.amazon.com/console/services?region=us-east-1

All Bookmarks

aws | Search [Alt+S] | Ask Amazon Q | United States (N. Virginia) | Vijay K (257527264832) | Vijay%20K

Console Home > All services

Console Home

- All services
- myApplications

All services

Services by category

Compute

- EC2
- Lightsail
- Lambda
- Batch
- Elastic Beanstalk
- Serverless Application Repository
- AWS Outposts
- EC2 Image Builder
- AWS App Runner
- Parallel Computing Service

Quantum Technologies

- Amazon Braket

Management & Governance

- AWS Organizations
- CloudWatch
- AWS Auto Scaling
- CloudFormation
- AWS Config
- Service Catalog

Security, Identity, & Compliance

- Resource Access Manager
- Cognito
- Secrets Manager
- GuardDuty
- Amazon Inspector
- Amazon Macie
- IAM Identity Center
- Certificate Manager
- Key Management Service
- CloudHSM

CloudShell | Actions

us-east-1

CloudShell | Feedback | Console Mobile App

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contd..

- **AWS Management Console** is a web application that comprises collection of **service-consoles** for **managing AWS resources**.
- **Console dashboard** appears ton log-in. It is starting point for interacting with various AWS services, other console components and is **single place** to access information needed to perform AWS related tasks.
- **Console Home page** customization is possible by adding, removing, and rearranging widgets, e.g. **recently visited**, **AWS Health etc**.
- **Individual service consoles** on the other hand offer a wide range of tools for cloud computing, as well as information about user account and about user billing.
- **AWS** is developing more widgets so user can further customize console experience. The dashboard consists of **a navigation bar** at the top and a number of **widgets** in the main body of the page, which user can configure and rearrange.

contd..

- The navigation bar at the top. In the image to the right, five controls within the navigation bar appear –

1. Account information
2. Region selector
3. Service selector
4. Search box
5. AWS Cloudshell

AWS Account information

- The first highlighted menu contains information and links for user account – **AWS Account ID**, **current user logged-in, links to navigate to the following pages** –

1. **Account** - account details viz.. user account, address, contact-info, etc.
2. **AWS Organization** - enables user to consolidate multiple AWS accounts into an organization thats created and centrally managed.
3. **Service Quotas - Quotas**, also referred to as limits in AWS services, are the maximum values for the resources, actions, and items in the AWS account. Default values assigned e.g. five Elastic IP addresses.
4. **Billing Dashboard** - page of the **AWS Billing** console is used to gain a general view of **user AWS spending**.
5. **Security Credentials** - This link takes user to AWS IAM user's page where user can **change password, add 2 factor authentication,** etc.
6. **Settings** - This link takes user to General Settings configuration page. User can manage console global settings such as default language and region as well preferences to optimize console display.

AWS Regions

- **AWS Region –** currently selected region is next highlighted menu
- **AWS global infrastructure** is grouped into **<u>Regions</u>**, and each service is hosted in a **Region**, unless its **global service**, such as IAM, or Amazon Route 53.
- The menu will display the currently selected Region, or "Global" when you have selected a global service. The text consists of the Region group, such as "US East"; region name, such as "N. Virginia"; and the Region string used by the AWS CLI, SDK, and other services, such as "us-east-1".
- When you click on the currently selected Region, a dropdown will appear with all the available Regions, and you can switch the console to a different Region by clicking on one in the list.
- Use the scroll bar on the right-hand side to see the full list of available options.

contd..

US East (N. Virginia)	us-east-1
US East (Ohio)	us-east-2
US West (N. California)	us-west-1
US West (Oregon)	us-west-2
Africa (Cape Town)	af-south-1
Asia Pacific (Hong Kong)	ap-east-1
Asia Pacific (Jakarta)	ap-southeast-3
Asia Pacific (Mumbai)	ap-south-1
Asia Pacific (Osaka)	ap-northeast-3
Asia Pacific (Seoul)	ap-northeast-2
Asia Pacific (Singapore)	ap-southeast-1
Asia Pacific (Sydney)	ap-southeast-2

aws Services Search for services, features, blogs, docs, and more [Option+S]

Recently visited

Favorites

All services

- Analytics
- Application Integration
- AR & VR
- AWS Cost Management
- Blockchain
- Business Applications
- Compute
- Containers
- Customer Enablement
- Database
- Developer Tools
- End User Computing
- Front-end Web & Mobile
- Game Development
- Internet of Things
- Machine Learning
- Management & Governance

Compute

AWS App Runner
Build and run production web applications at scale

Batch
Fully managed batch processing at any scale

EC2
Virtual Servers in the Cloud

EC2 Image Builder
A managed service to automate build, customize and deploy OS images

Elastic Beanstalk
Run and Manage Web Apps

Lambda
Run Code without Thinking about Servers

Lightsail
Launch and Manage Virtual Private Servers

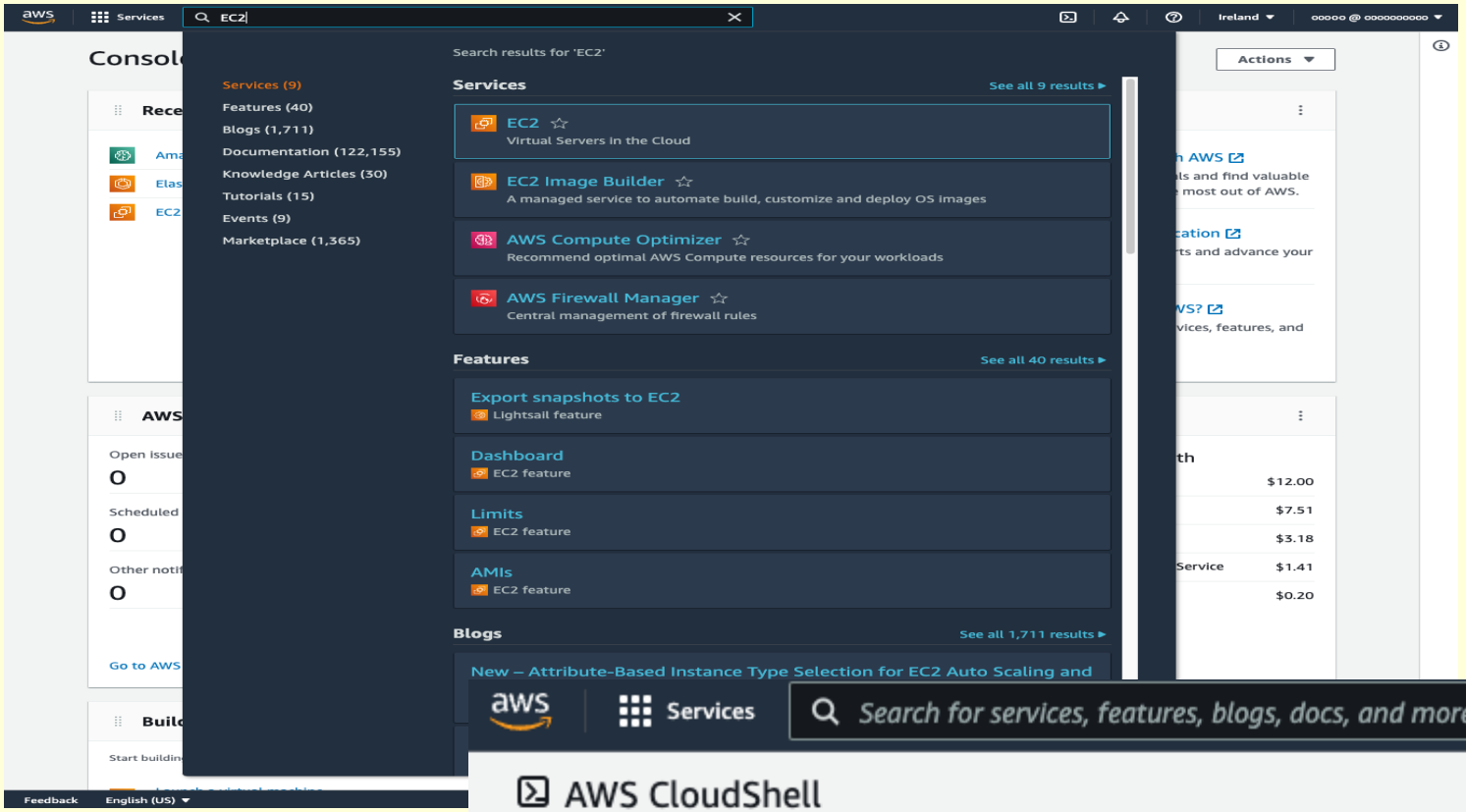
AWS Outposts
Run AWS Services On Premises

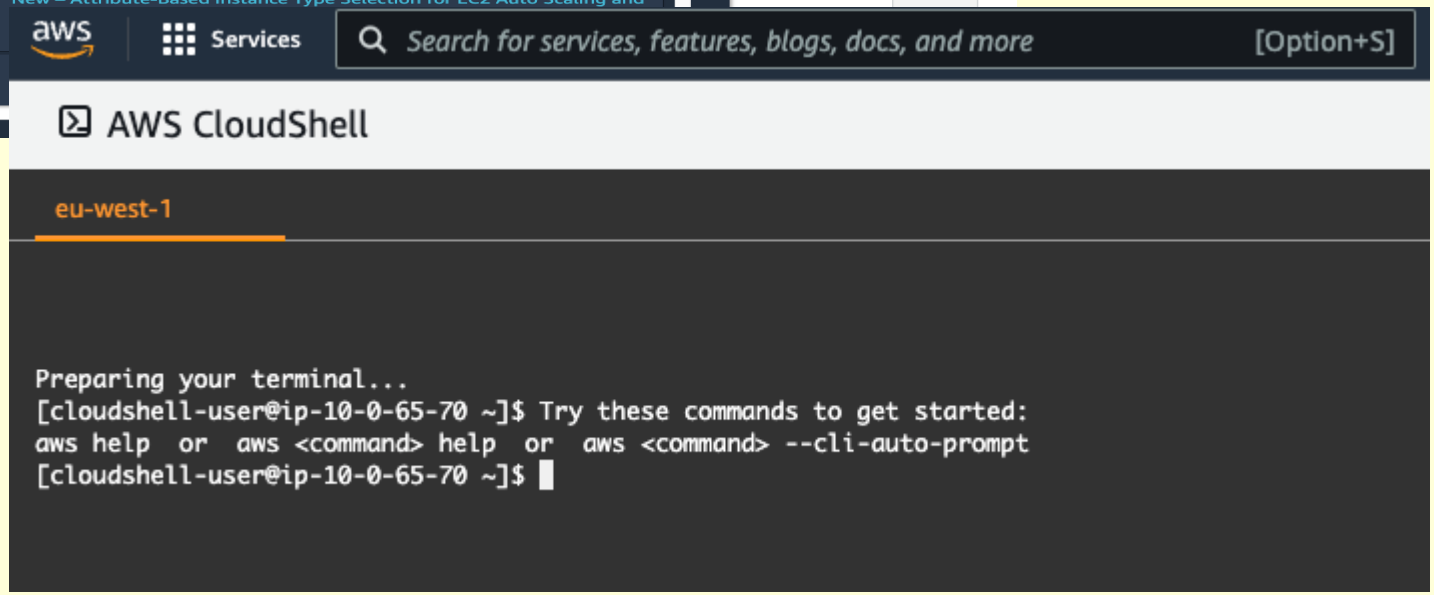
Serverless Application Repository
Assemble, deploy, and share serverless applications within teams or publicly

AWS CloudShell

- The fifth highlighted item is the AWS CloudShell icon.
- This icon will launch a browser-based shell environment that is pre-authenticated with your console credentials.
- Use this to execute AWS CLI commands or scripts using the AWS CDK from your browser.
- If you add any files to CloudShell (up to the 1GB limit), it will persist the files between sessions.
- CloudShell is a useful tool for securely interacting with AWS account, and is Region specific, so any files that you upload are specific to that Region.
- The current selected Region is displayed in orange above the terminal.

contd..


aws
Services
EC2
Ireland
Actions
Search results for 'EC2'
Services (9)
Features (40)
Blogs (1,711)
Documentation (122,155)
Knowledge Articles (30)
Tutorials (15)
Events (9)
Marketplace (1,365)
Services
See all 9 results
EC2
Virtual Servers in the Cloud
EC2 Image Builder
A managed service to automate build, customize and deploy OS images
AWS Compute Optimizer
Recommend optimal AWS Compute resources for your workloads
AWS Firewall Manager
Central management of firewall rules
Features
See all 40 results
Export snapshots to EC2
Lightsail feature
Dashboard
EC2 feature
Limits
EC2 feature
AMIs
EC2 feature
Blogs
See all 1,711 results
New – Attribute-Based Instance Type Selection for EC2 Auto Scaling and
Feedback
English (US)


aws
Services
Search for services, features, blogs, docs, and more
[Option+S]
AWS CloudShell
eu-west-1
Preparing your terminal...
[cloudshell-user@ip-10-0-65-70 ~]$ Try these commands to get started:
aws help or aws <command> help or aws <command> --cli-auto-prompt
[cloudshell-user@ip-10-0-65-70 ~]$

AWS Cost Management

File Edit View History Bookmarks Tools Help

Cloud Computing Services - Am

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aws

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Products Solutions Pricing Documentation Learn Partner Network AWS Marketplace Customer Enablement Ev

- AWS Management Console
- Account Settings
- Billing & Cost Management
- Security Credentials
- AWS Personal Health Dashboard

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Cost Management

https://us-east-1.console.aws.amazon.com/cost-management/home?region=us-east-1#/startu

Search

aws Services Search for services, features, blogs, docs, and more [Alt+S] Global Vijay

AWS Cost Management

- Home
- Cost Explorer
- Reports
- Budgets
- Cost Anomaly Detection
- Rightsizing recommendations

Savings Plans

- Overview
- Inventory
- Recommendations
- Purchase Savings Plans
- Utilization report
- Coverage report

Welcome to AWS Cost Management

Since this is your first visit, it will take some time to prepare your cost and usage data. Please check back in 24 hours.

Feedback Looking for language selection? Find it in the new Unified Settings

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AWS Search

- The fourth highlighted item is the **search box**. When you enter text into the box, the underlying search engine searches across a number of different locations to match the text that you entered. It will return matches in eight sections:
- If the search engine is unable to find a match within one of these sections, the section will not be included in the list.
- **Services**: List of AWS services
- **Features**: List of features of AWS services
- **Blogs**: Posts from the AWS blog
- **Documentation**: AWS Documentation
- **Knowledge Articles**: AWS Premium Support Knowledge Center
- **Tutorials**: Hands-on guides from the AWS Getting Started Resource Center
- **Events**: AWS hosted events that are upcoming, or available on-demand
- **Marketplace**: AWS Marketplace offerings that you can deploy in your AWS account.

AWS Dashboard Widgets

- The dashboard also contains a number of widgets to help get started. Three of the widgets provide static links to learn how to build a solution or to explore AWS (Welcome to AWS, Build a solution, and Explore AWS).
- The other five are dynamic and provide important information about AWS services, AWS costs and usage, and best practices:
- **AWS Health**: information on events that might affect your AWS infrastructure and account
- **Cost and usage**: an overview of service costs, with a breakdown by AWS service
- **Favorites**: a list of your favorite AWS services
- **Recently visited**: a list of top recently visited services
- **Trusted Advisor**: recommendations to follow AWS best practices